Project Specification: Mini Spotify

Object-Oriented Programming (Java) — Music and Podcast Management System

You are required to develop an Object-Oriented Programming (OOP) application in Java for a digital audio platform. The application is a simplified content management system capable of handling various content types (Songs, Podcasts), managing user Playlists, and maintaining Artist catalogs.

REQUIRED OOP CONCEPTS

- Classes, Objects & Constructors
- Encapsulation (Access Modifiers)
- `this` & `super` Keywords
- `static` & `final` Modifiers
- Inheritance (Base & Subclasses)
- Method Overloading & Overriding
- Polymorphism & Dynamic Binding
- Type Casting & `instanceof` Operator
- `toString()` & `equals()` Overriding
- Collection Handling (`ArrayList`)

UML DIAGRAM NOTATION LEGEND

`-` Private field	`+` Public member	`#` Protected field
Underlined Static / Final member	Δ Inheritance (is-a)	♦ Composition (contains)

CORE CLASS SPECIFICATIONS

1. Content (Base Class — *Icerik*) — Abstract / Parent Class

Attributes:

- `- contentId : int (icerikNo)`
- `- title : String (baslik)`
- `- duration : int (sure - seconds)`
- `- playCount : int (oynatilmaSayisi)`
- `- contentCount : int {static} (icerikSayisi)`
- `+ PLATFORM_NAME : String {static, final} (PLATFORM_ADI)`

Key Methods:

- `+ Content(contentId : int, title : String, duration : int)`
- `+ play() : void` — Increments `playCount` and outputs playing status
- `+ equals(obj : Object) : boolean & + toString() : String`
- `+ getContentCount() : int {static}`
- `+ displayPlatformInfo() : void {final}`

2. Song (Subclass of Content — *Sarki*) — Inherits Content

Attributes:

- `- artist : String (sanatci)`
- `- album : String (album)`
- `- genre : String (tur)`

Key Methods:

- `+ Song(contentId, title, duration, artist, album, genre)`
- `+ play() : void {override}` — Outputs song metadata during playback

CLASS SPECIFICATIONS (CONTINUED)

3. Podcast (Subclass of Content — *Podcast*)

Inherits Content

Attributes:

- `- host : String (sunucu)`
- `- episodeNumber : int (bolumNo)`

Key Methods:

- `+ Podcast(contentId, title, duration, host, episodeNumber)`
- `+ play() : void {override}` — Outputs host and episode details
- `+ toString() : String {override}`

4. Playlist (*CalmaListesi*)

Composition (0..* Content)

Attributes:

- `- playlistName : String (listeAdi)`
- `- contents : ArrayList<Content> (icerikler)`

Key Methods:

- `+ Playlist(playlistName : String)`
- `+ addContent(content : Content) : void (icerikEkle)`
- `+ addContent(content : Content, showInfo : boolean) : void {overload}`
- `+ removeContent(content : Content) : boolean (icerikSil)`
- `+ displayList() : void & + playList() : void`
- `+ calculateTotalDuration() : int` — Returns total seconds in playlist
- `+ findLongestContent() : Content & + findMostPlayed() : Content`
- `+ listSongsByArtist(artist : String) : void`
- `+ listOnlySongs() : void & + listOnlyPodcasts() : void (uses instanceof)`

5. User (*Kullanici*)

Owns (0..* Playlist)

Attributes:

- `- username : String (kullaniciAdi)`
- `- fullName : String (adSoyad)`
- `- isPremium : boolean (premium)`
- `- playlists : ArrayList<Playlist> (calmalisteleri)`

Key Methods:

- `+ User(username : String, fullName : String, isPremium : boolean)`
- `+ createPlaylist(playlistName : String) : Playlist`
- `+ getPlaylists() : ArrayList<Playlist> & + setPremium(isPremium : boolean)`

6. Artist (*Sanatci*)

Association (0..* Song)

Attributes:

- `- artistName : String (sanatciAdi)`
- `- genre : String (tur)`
- `- songs : ArrayList<Song> (sarkilar)`

Key Methods:

- `+ Artist(artistName : String, genre : String)`
- `+ addSong(song : Song) : void (sarkiEkle)`
- `+ getSongs() : ArrayList<Song>`